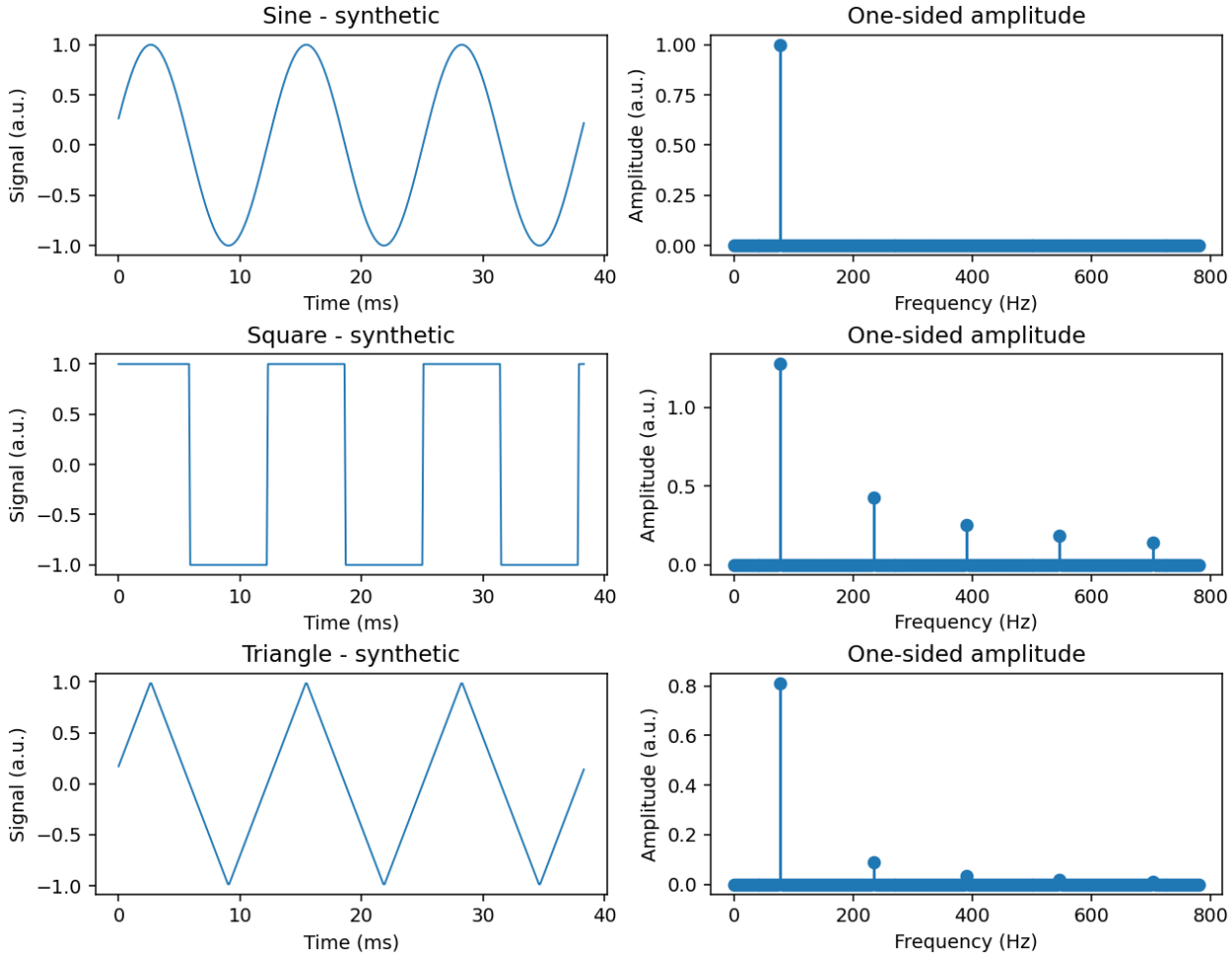
Laboratory 2 | Fourier analysis

Reproducible reconstruction of the University of Pisa FFT exercise

Data provenance. All records and figures are synthetic. Original Arduino measurements were not provided. This report gives the theoretical expectations and demonstrates the analysis; it does not report experimental measurements.

1. Periodic waveforms

With 2048 samples separated by 100 microseconds, the sampling frequency is 10,000 Hz, the Nyquist frequency is 5,000 Hz, and the bin spacing is 4.8828125 Hz. The fundamental at 78.125 Hz completes exactly 16 cycles in the 0.2048 s DFT record, avoiding partial-cycle leakage.



Unit-amplitude synthetic signals and their one-sided FFT amplitudes.

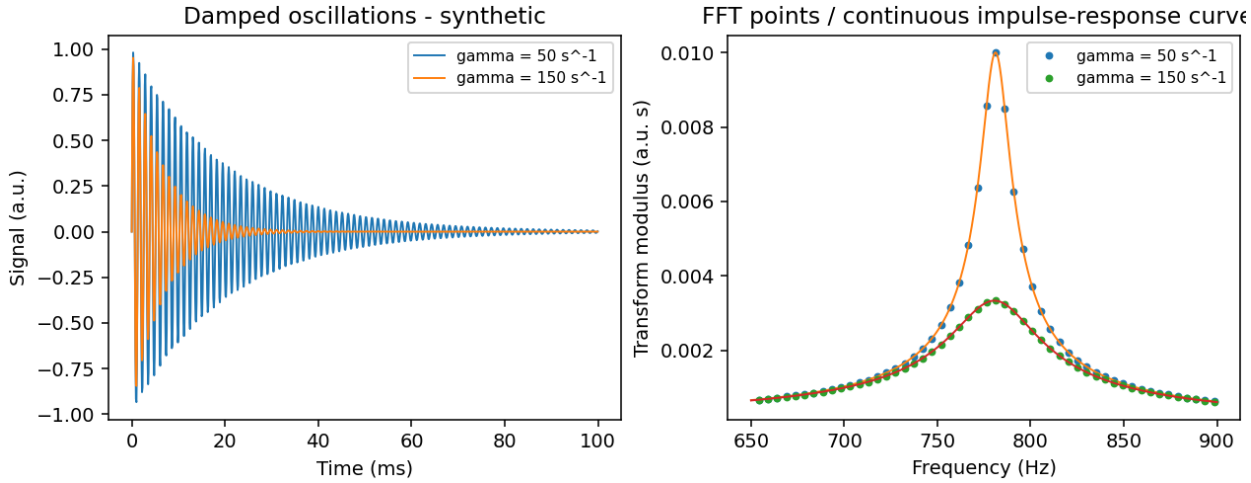
A sine gives one spectral line. Square and triangular waves have odd harmonics, with relative amplitudes 1/h and $1/h^2$. The triangle's smoother waveform needs less high-frequency content.

h	Square FFT	Theory	Triangle FFT	Theory
3	0.333601	0.333333	0.111022	0.111111
5	0.200483	0.200000	0.039903	0.040000
7	0.143548	0.142857	0.020309	0.020408
9	0.112009	0.111111	0.012245	0.012346

2. Damped oscillator

We use x(t) = exp(-gamma t) sin(omega_d t) for t >= 0, where omega_d^2 = omega_0^2 - gamma^2. The natural frequency is fixed at 781.25 Hz; gamma is either 50 or 150 s^{-1}.

Its continuous transform is omega_d / [omega_0^2 - (2 pi f)^2 + 4 pi i gamma f]. This follows by integrating the two decaying complex exponentials that make up the sine.



FFT points approximate the continuous impulse-response transform (lines).

Interpretation. Stronger damping shortens the train of oscillations. Fewer coherent cycles give a broader frequency peak. With the same initial amplitude, the integrated response is smaller, so its peak is lower.

Near resonance, for weak damping, the power spectrum has full width at half maximum approximately gamma/pi: 15.9155 and 47.7465 Hz. The modulus has a different half-height width, approximately sqrt(3) gamma/pi. The quality factor is omega_0/(2 gamma).

gamma = 50 s^{-1}: FFT peak bin 781.2500 Hz; continuous peak 781.1689 Hz; maximum complex FFT error relative to the exact discrete reference 5.30e-15.

gamma = 150 s^{-1}: FFT peak bin 781.2500 Hz; continuous peak 780.5201 Hz; maximum complex FFT error relative to the exact discrete reference 2.36e-15.

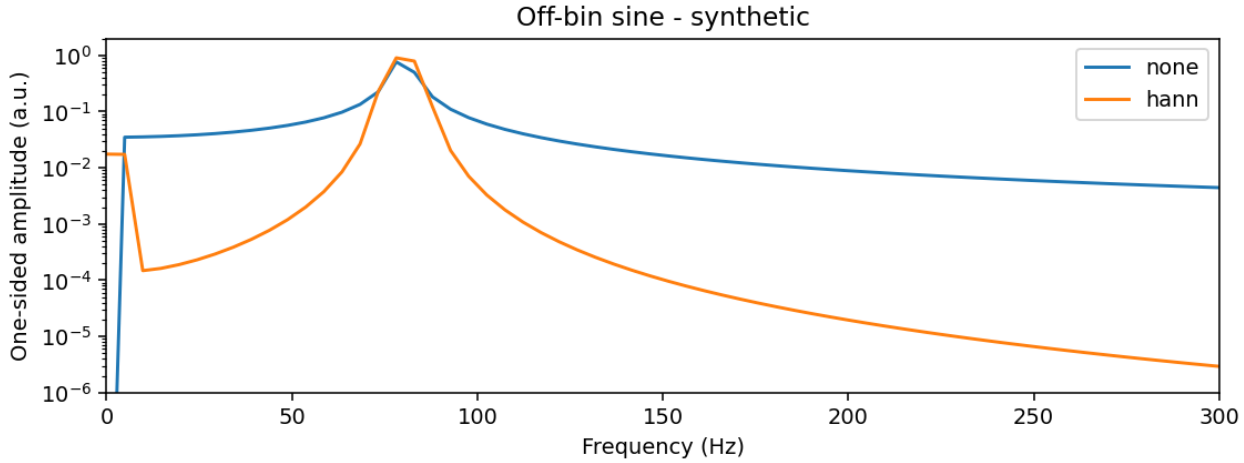
The finite sampled transform is independently evaluated using geometric sums. The continuum peak is sqrt(omega_0^2 - 2 gamma^2)/(2 pi), slightly different from the ringdown frequency. The 4.8828125 Hz FFT grid cannot show that small shift directly.

Connection to RLC. The transform of an impulse response is proportional to the corresponding transfer function. This comparison assumes the appropriate initial conditions and observed variable. Capacitor and resistor voltages, or ringdowns with a different initial phase, need not have the same numerator.

For the transient we plot dt times |FFT|, in signal units times seconds, without mean subtraction or windowing. This is different from the periodic waveforms' single-sided peak amplitudes.

3. Finite records and experimental data

A pure sine completing 16.4 cycles does not join smoothly across the ends of the record. Its spectral weight spreads over neighbouring bins: this is spectral leakage. A Hann window reduces distant tails while broadening the central peak. Broadening alone therefore does not prove physical distortion.



The same synthetic off-bin sine with and without a Hann window.

Normalization. For an unwindowed record of N samples, the one-sided amplitude is 2|FFT|/N at interior positive frequencies. DC and, for even N, Nyquist are not doubled. With Hann weighting the denominator is the sum of the weights. This corrects the amplitude of an isolated bin-centred tone, but not the peak-bin amplitude of every off-bin tone.

Sampling. Frequencies above Nyquist fold into the observed band (aliasing). Ideal square and triangular waves are not band-limited, which explains small sampling differences in the harmonic table. NumPy does not require a power-of-two record length. Zero padding adds plotting points rather than new frequency information.

Real recordings. analyze.py reads two-column time/signal records and converts microseconds, milliseconds, or seconds. It checks strictly increasing finite timestamps. Deviations above 1% of the mean sample interval require explicit opt-in. This threshold is only a practical warning: accepting the mean interval does not correct timing jitter.

The DFT duration is N dt; the first-to-last timestamp interval is (N-1)dt. A larger sampling rate extends the observable band; a longer observation improves frequency separation. Bin spacing is not a statistical error bar.

Reproduce. Run demo.py, build_report.py, and python -m unittest discover -s tests -v. See README.md for installation and recording-analysis commands; solution.md contains the full explanation.

Source exercise: Francesco Fuso, FFT exercises, University of Pisa (12 April 2017).

Implementation references: NumPy rfft and rfftfreq.